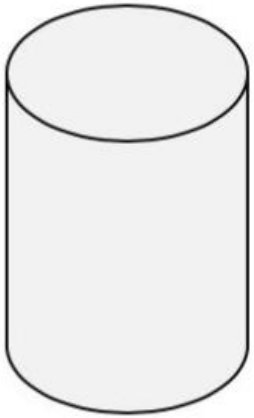
Data Warehouse Design

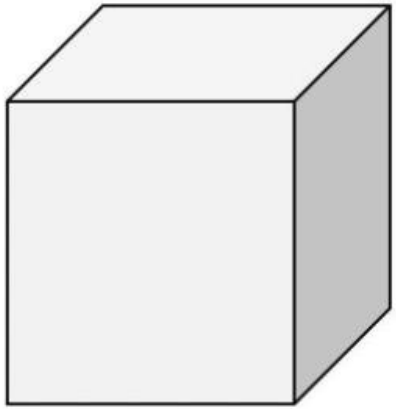
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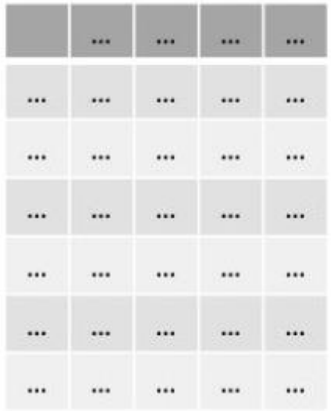
- Taniar, D & Rahayu, W 2021, Data Warehousing and Analytics: Fueling the Data Engine. Data-Centric Systems and Applications, 1st edn, Springer, Cham Switzerland. https://doi.org/10.1007/978-3-030-81979-8
- Haryanto, A 2022, FIT3003 – Business Intelligence and Data Warehousing, Monash University, Australia

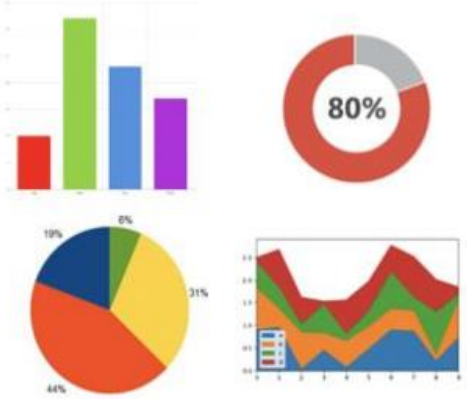
Recall – The Big Picture

Operational Database → Data Warehouse → OLAP → Business Intelligence

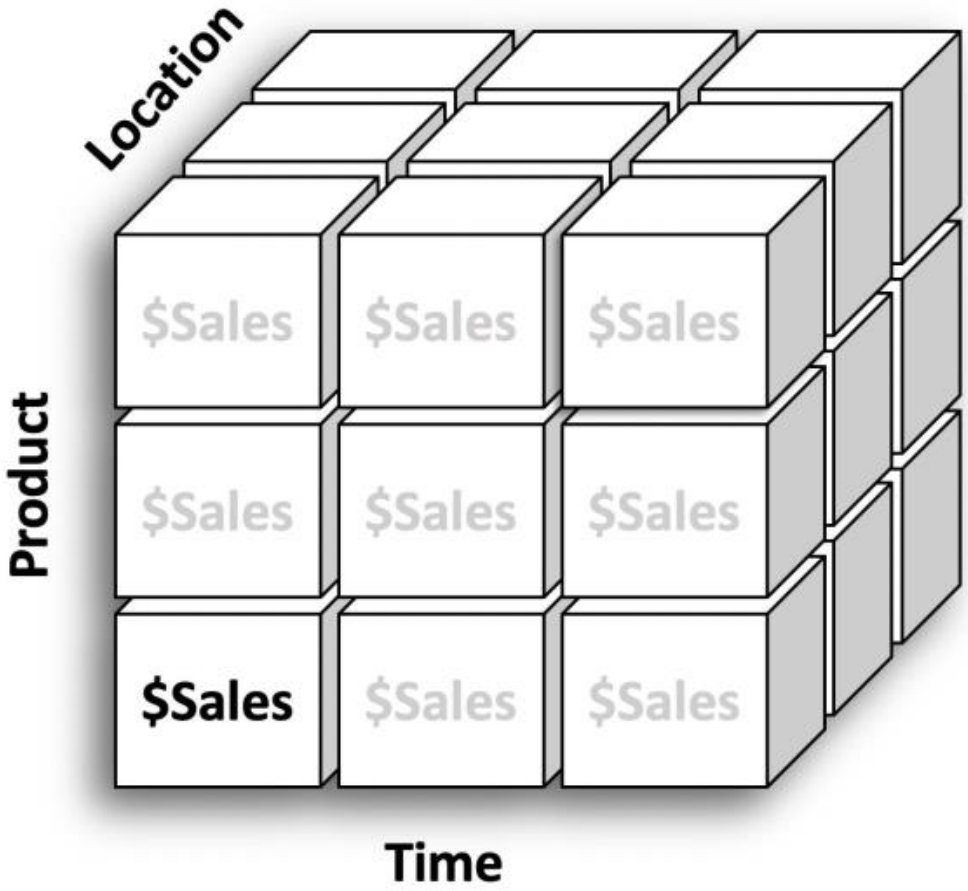








Recall – Data Warehouse



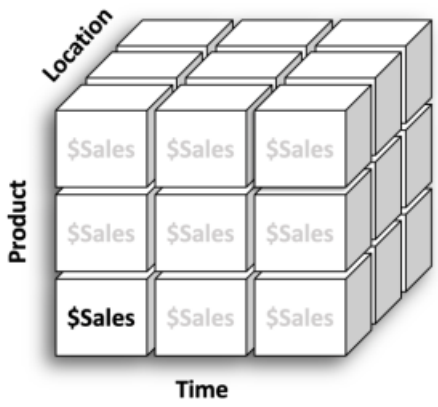
Star Schema

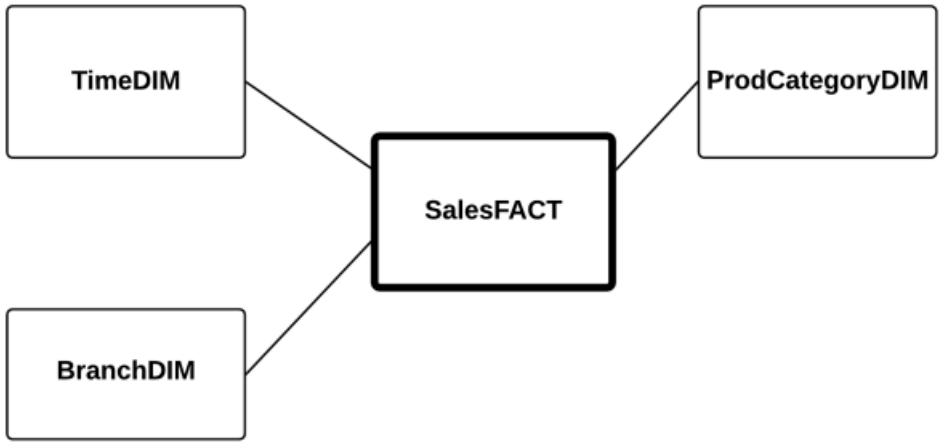
- A Star Schema is a design representation of a multi-dimensional view. It is a data modeling technique used to map multidimensional decision support data into a relational database.

- The reason for the star schema's development is that existing relational modeling techniques: ER and normalization, did not yield a database structure that served the advanced data analysis requirements well.

Star Schema Components

- There are **Three** main components of the Star Schema:
  1. Facts
  2. Dimensions
  3. Attributes





Star Schema Components

1. **Facts**

   Facts are **numeric measurements** (values) that represent a specific business aspect or activity.

   For example, sales figures are numeric measurements that represent product and/or service sales.

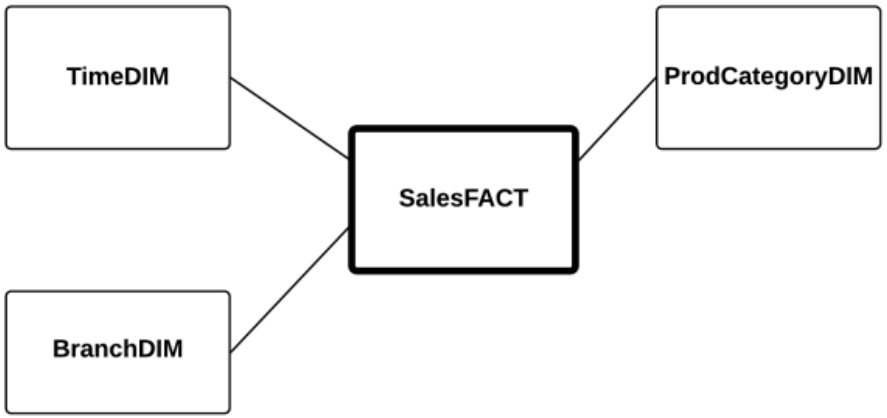
2. **Dimensions**

   Dimensions are qualifying characteristics that provide **additional perspectives** to a given fact.

   For example, sales might be *viewed* from specific dimension(s), such as sales location, sales period, sales product, etc.

Star Schema Notation

- A Sales Star Schema
  - **Fact**:
    - Sales
  - **Dimensions**:
    - Time
    - Product
    - Branch



- Notation-wise, the Fact uses a bolder line, to differentiate between Fact from Dimensions.

Star Schema Components

3. Attributes

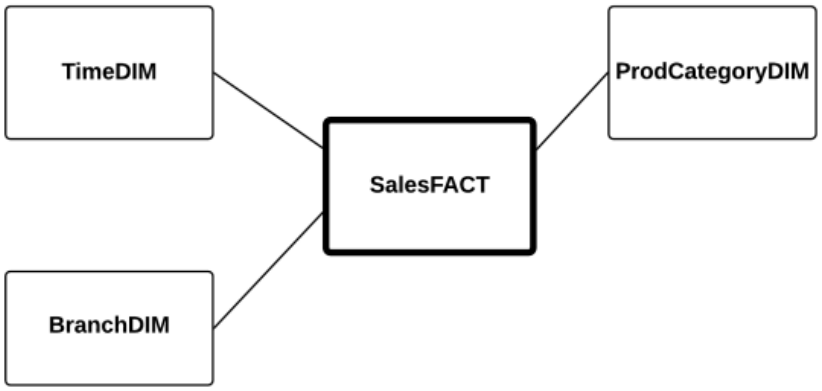
Each dimension table contains attributes.

For example:

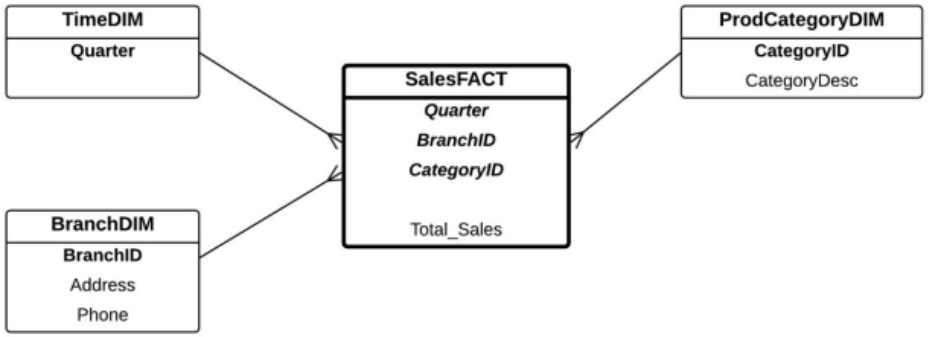
Product dimension:	Prod Type, Description.
Location dimension:	Region, State, City.
Time dimension:	Year, Month.

Star Schema Notation

- Sales Star Schema



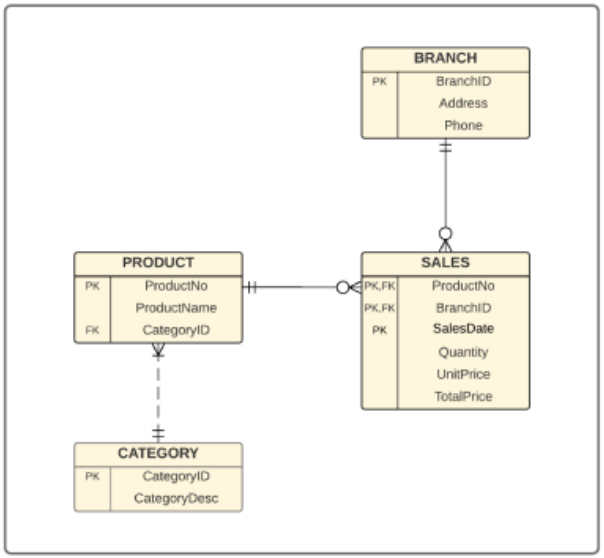
(a) Outline star schema for Sales



(b) Sales Star Schema complete with the Attributes

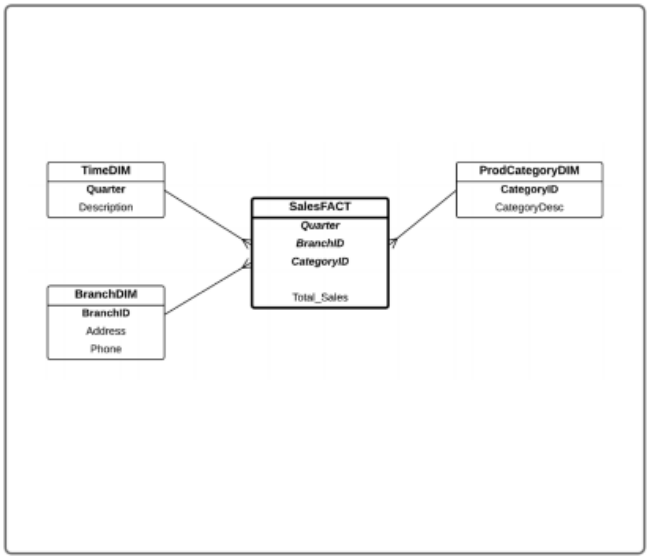
Transformation Process

Operational Database (E/R Diagram)



Transformation (ETL)

Data Warehouse (Star Schema)



Case Study #1: International College

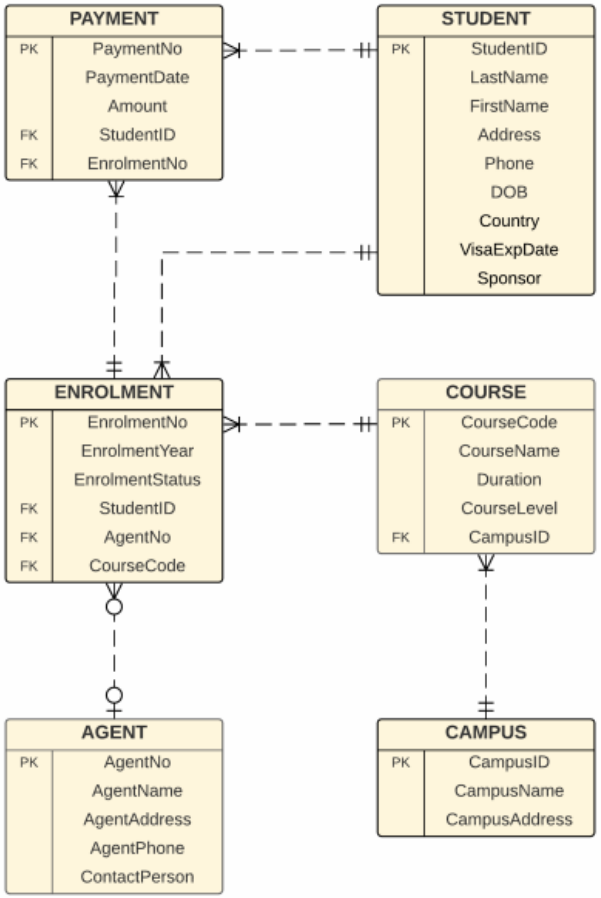
The admission office handles enrolment, payment, and marketing campaigns to international students, often through educational agents located overseas. This admission office has an operational system that maintains all the details of international students enrolled in the College. Payment details are also handled by this office. Basically, the operational system has the following features:

- Every student details are kept in the database. This includes the courses that the students enroll.
- As the College is a multi-campus university, some courses are offered in a different campus. The admission office handles international students of all campuses.
- Some international students coming to the College are handled by an educational agent. This is particularly common for the first course that a student enrolls in. Subsequent courses are not normally handled by an agent, because the students themselves deal directly with the College.
- International students pay tuition fees several times (normally once every semester) for each course they are doing.

Case Study #1: International College

The College now requires a data warehouse for analysis purposes. The analysis is needed for identifying at least the following questions:

1. How many students come from certain countries?
2. What is the total income for certain postgraduate courses?
3. How many students are handled by certain agents?
4. How the number of enrolment of courses fluctuates across the year?

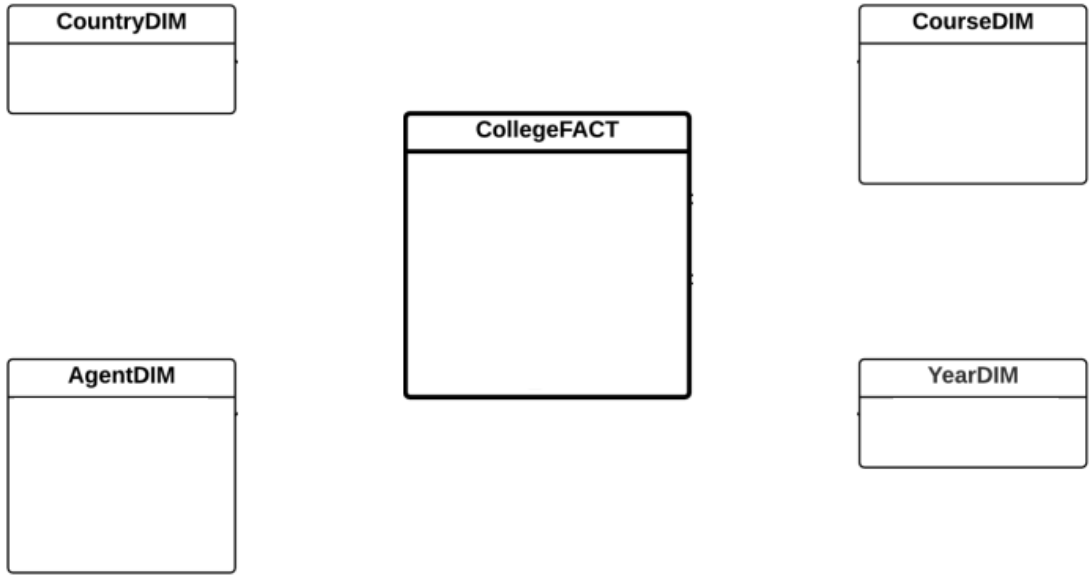


Case Study #1: International College

- College Star Schema
  - **Fact**:
    - Number of Students
    - Total Income
  - **Dimensions**:
    - Country
    - Agent
    - Course
    - Year

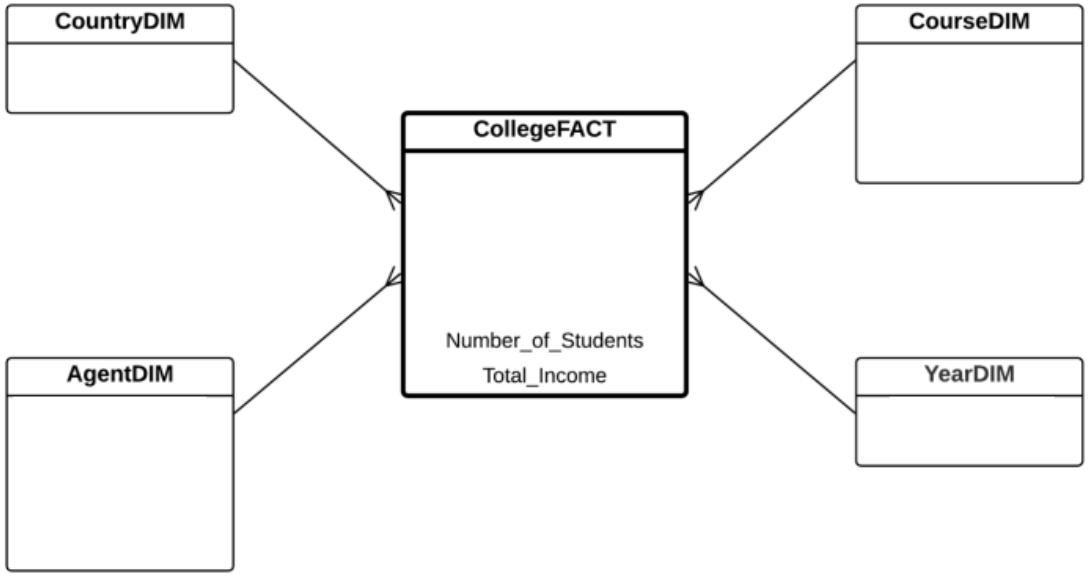
Case Study #1: International College

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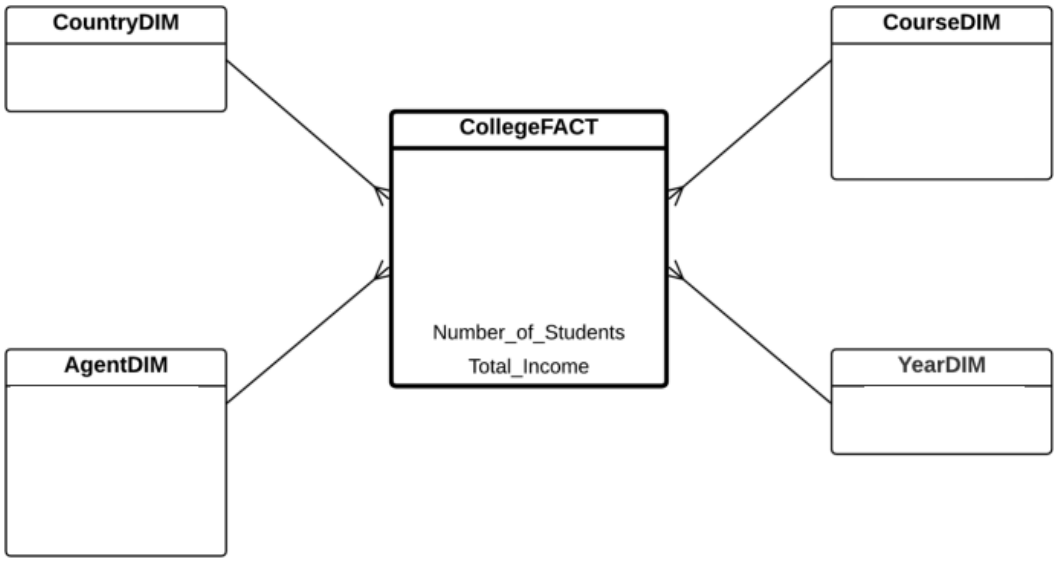


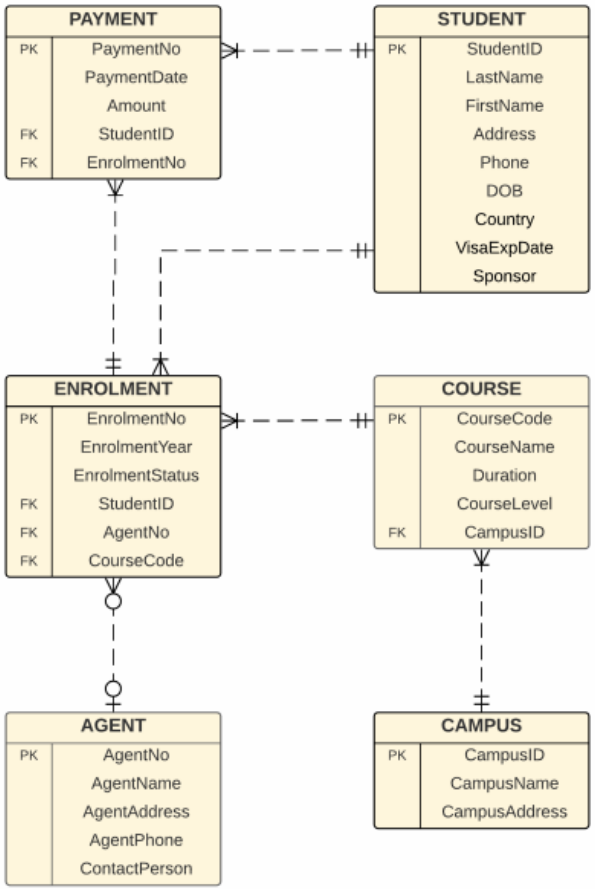
Case Study #1: International College

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    - Number of Students
    - Total Income
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    - Country
    - Agent
    - Course
    - Year

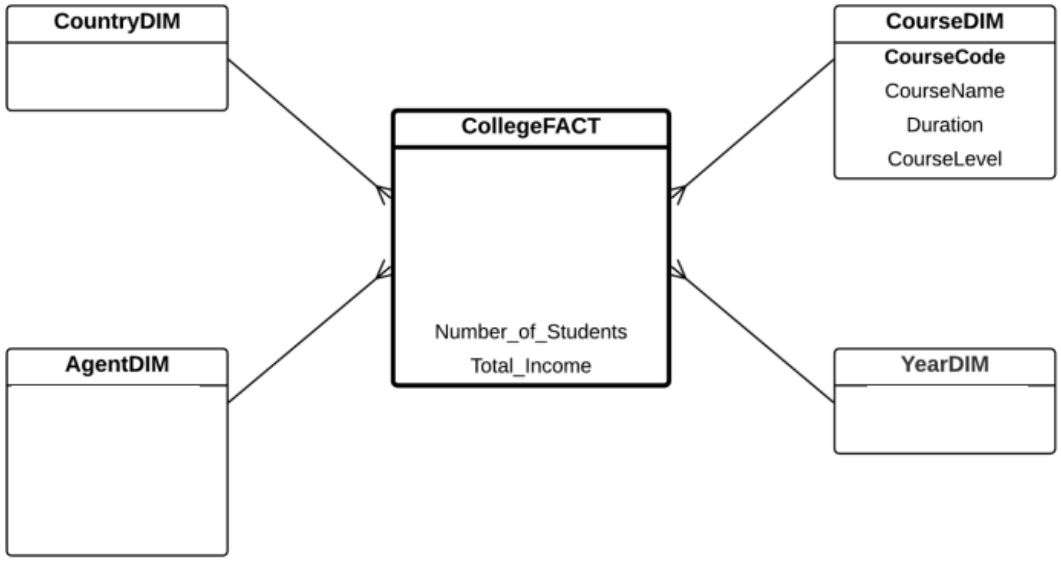


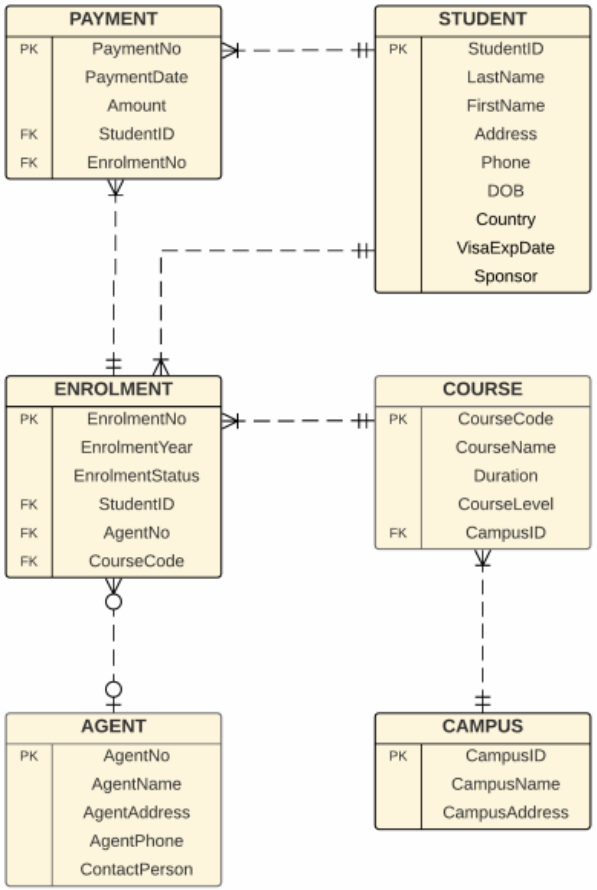
Case Study #1: International College



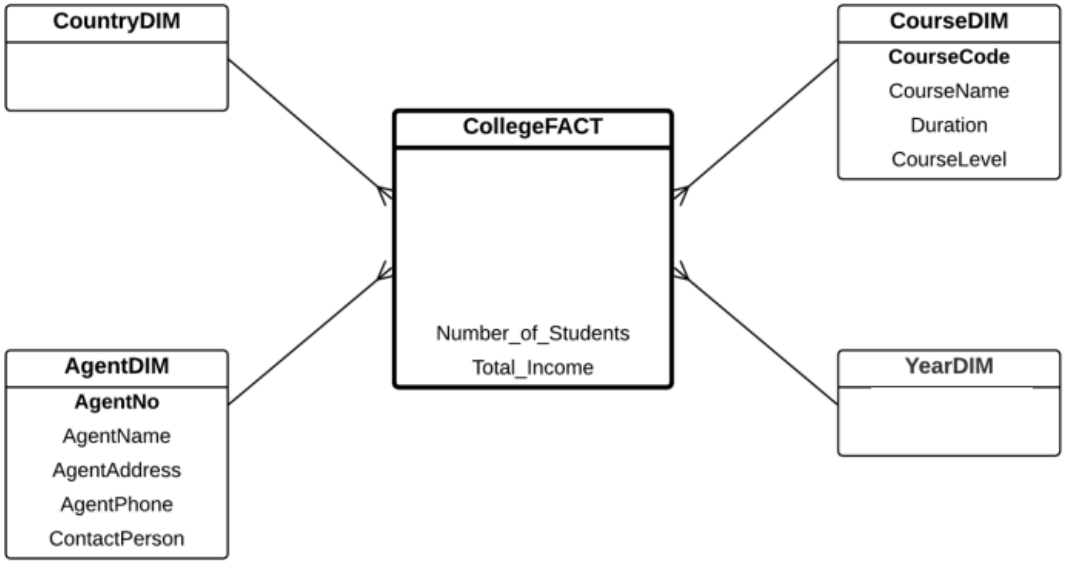


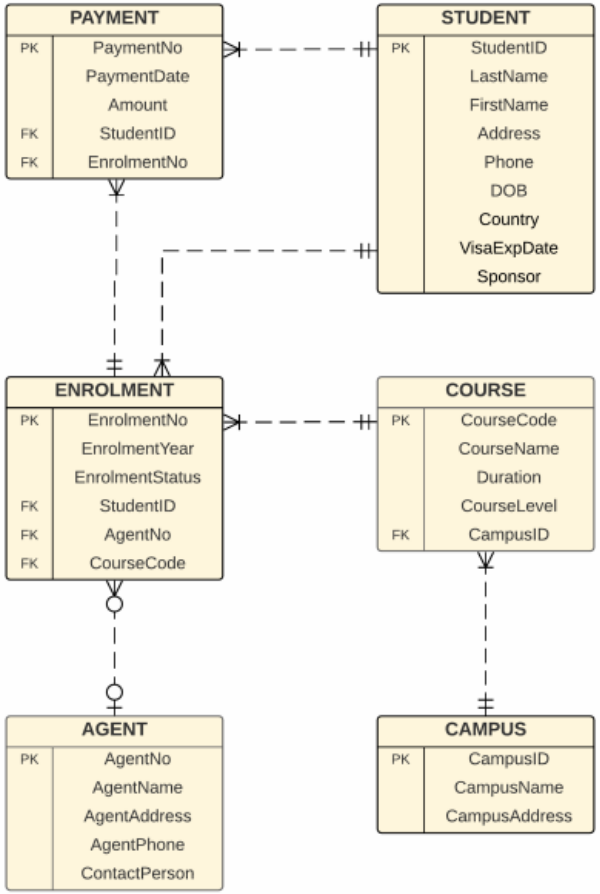
Case Study #1: International College



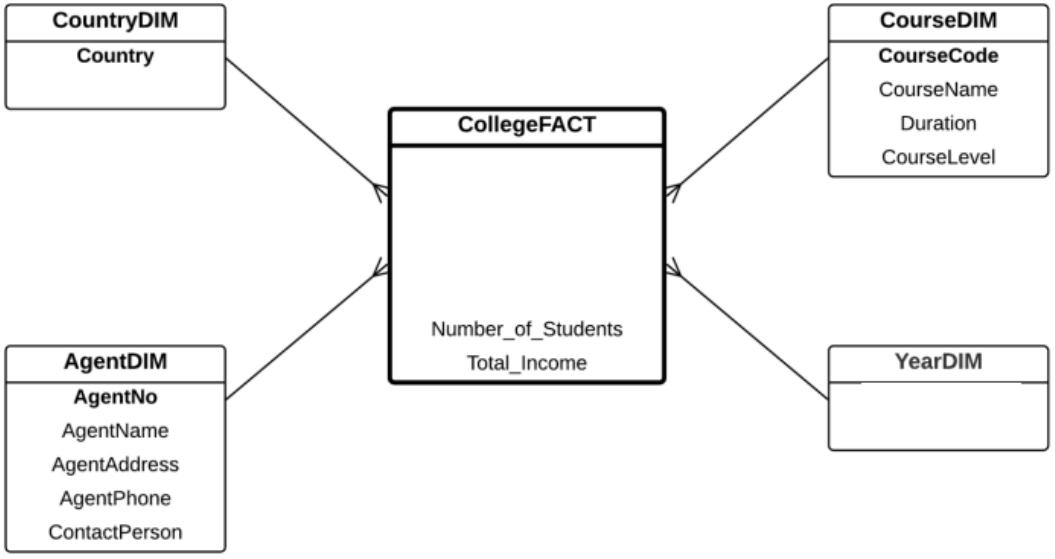


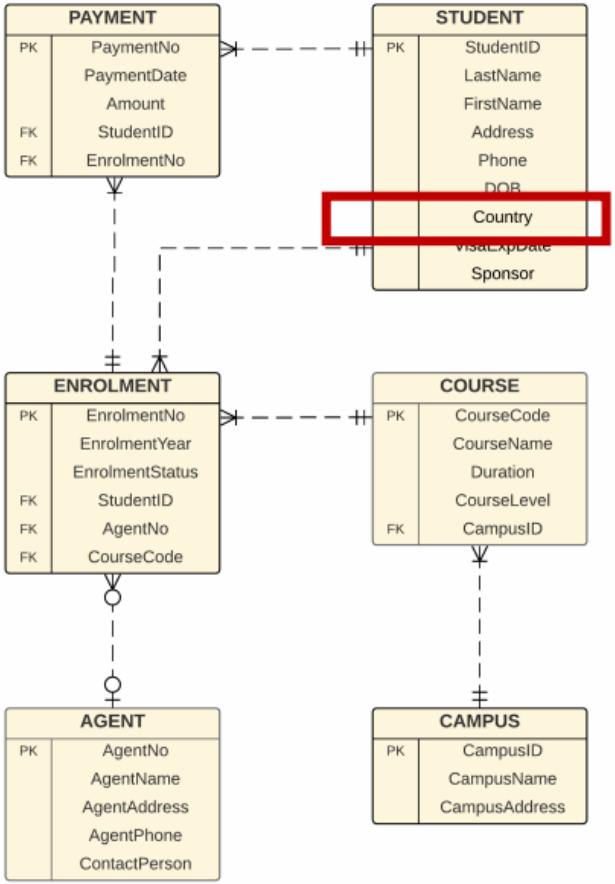
Case Study #1: International College



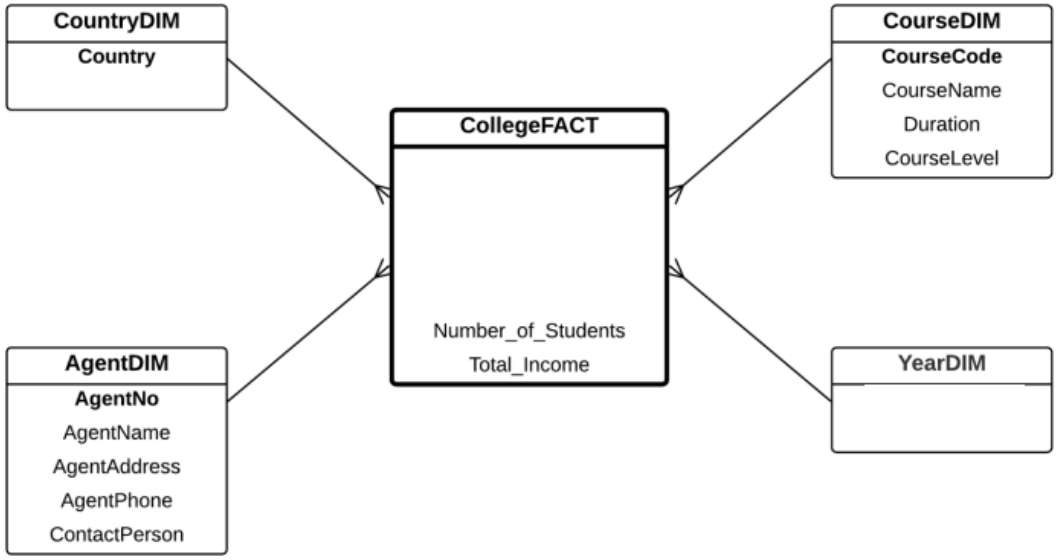


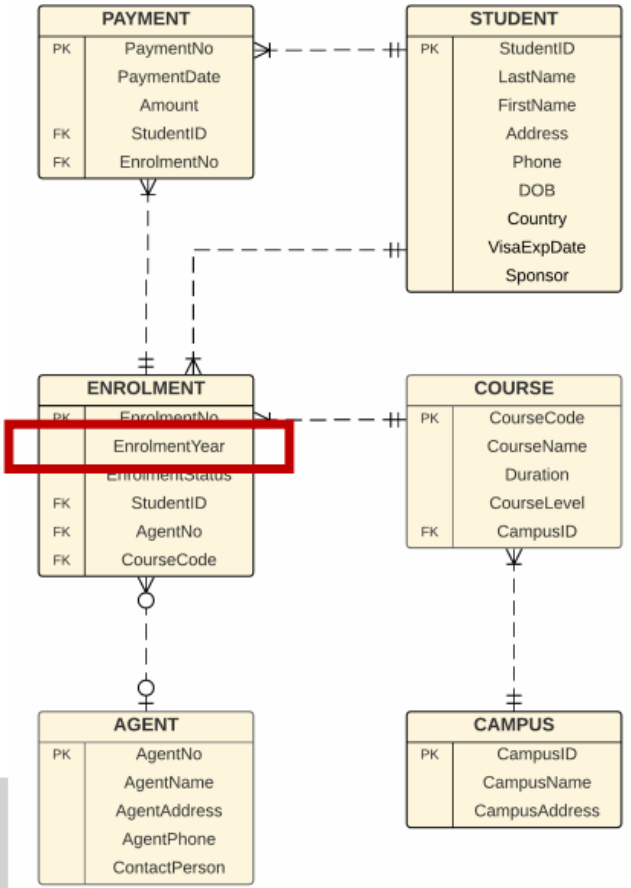
Case Study #1: International College



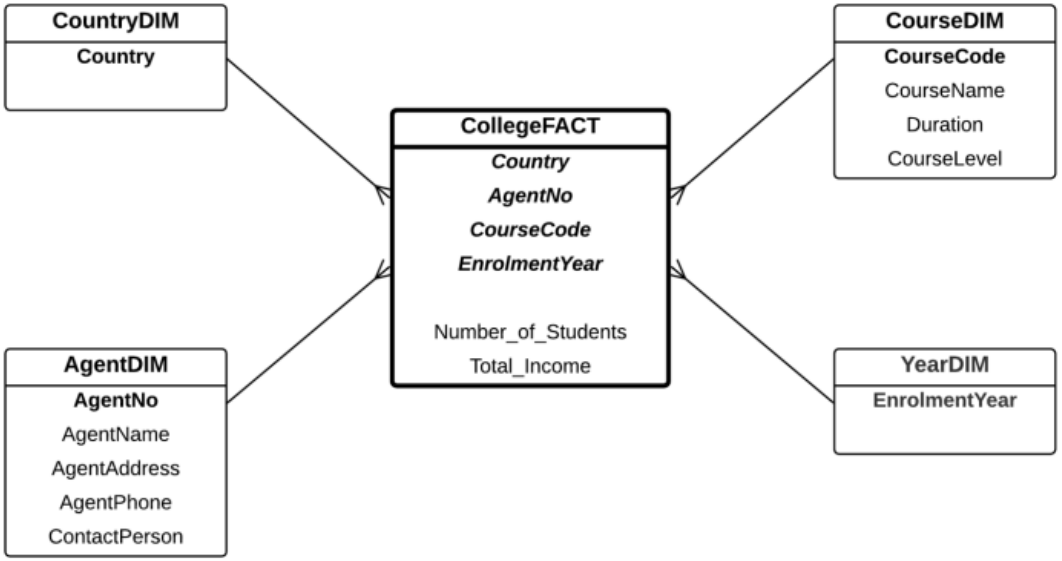


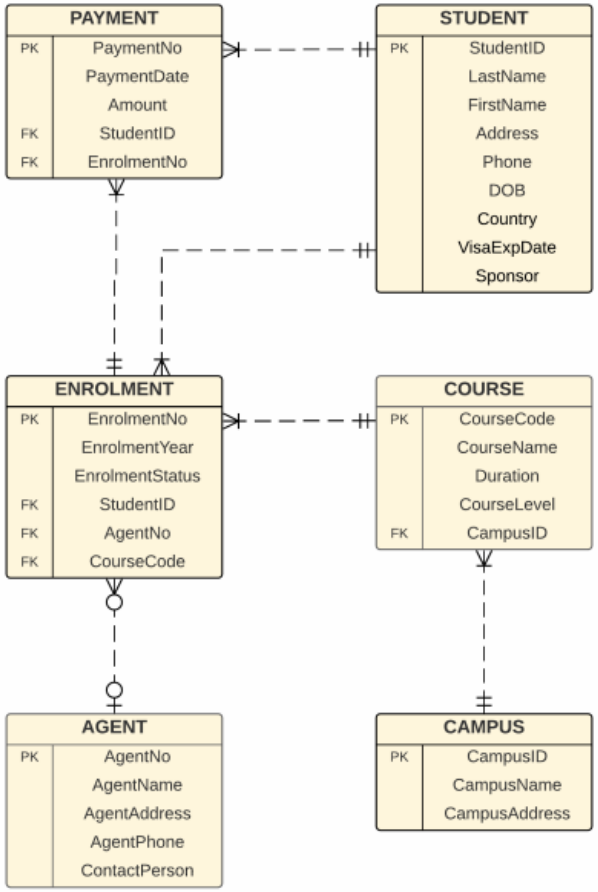
Case Study #1: International College





Case Study #1: International College

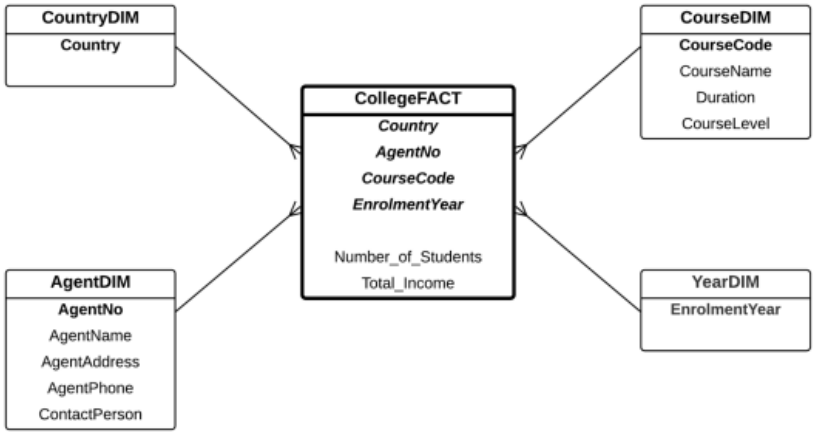




Case Study #1: International College

- To create AgentDIM:
  - `create table AgentDim as select * from Agent;`

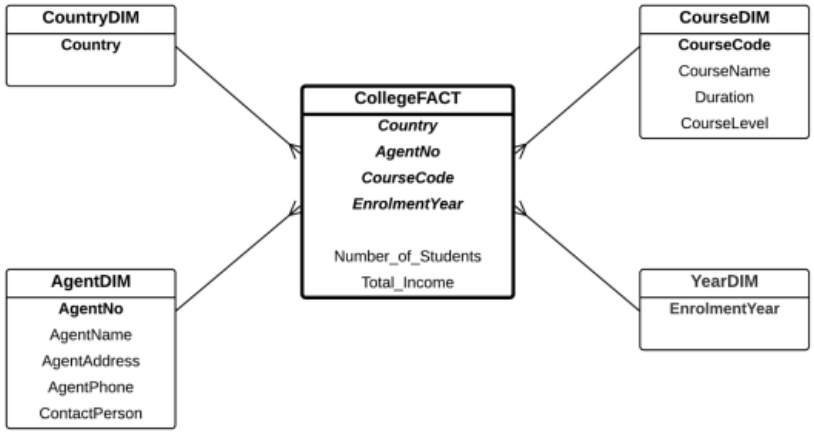
- To create CountryDIM:
  - `create table CountryDim as select distinct Country from Student;`



Case Study #1: International College

- To create CourseDIM:
  - `create table CourseDim as`
    `select CourseCode, CourseName, Duration, CourseLevel`
    `from Course;`

- To create YearDIM:
  - `create table YearDim as`
    `select distinct EnrolmentYear`
    `from Enrolment;`

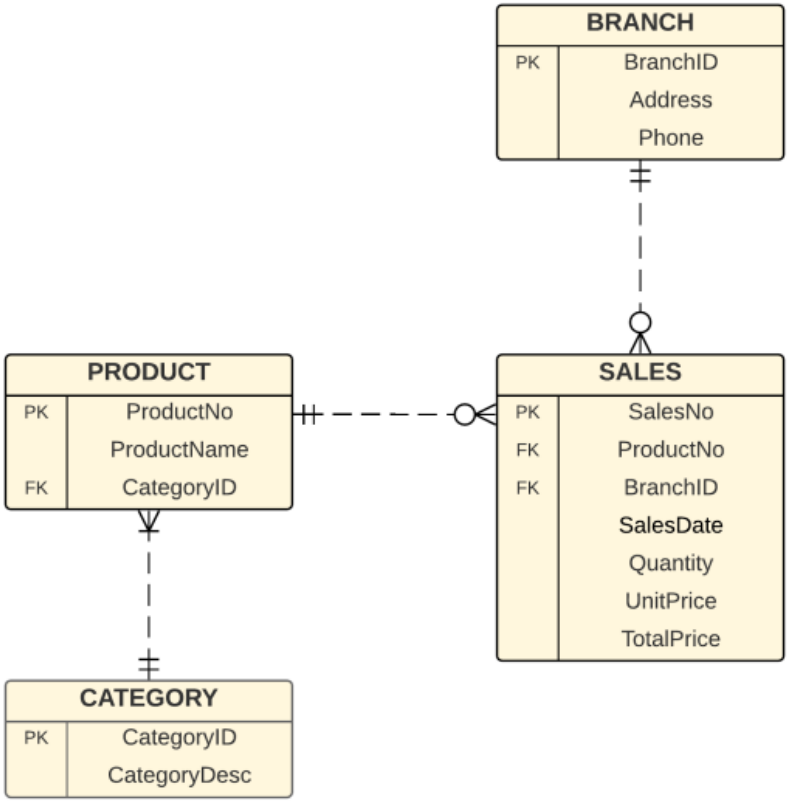


Case Study #1: International College

- To create CollegeFACT:
  -
```
create table CollegeFact as
Select S.Country, E.AgentNo, E.CourseCode, E.EnrolmentYear,
count(S.StudentID) as Number_of_Students,
sum(P.Amount) as Total_Income
from Student S, Enrolment E, Payment P
where E.EnrolmentNo = P.EnrolmentNo
and E.StudentID = S.StudentID
group by S.Country, E.AgentNo, E.CourseCode, E.EnrolmentYear;
```

Case Study #2: Sales

- Suppose that we would like to analyze Total Sales from various point of views, such as Quarter, Branch, and Product Category.

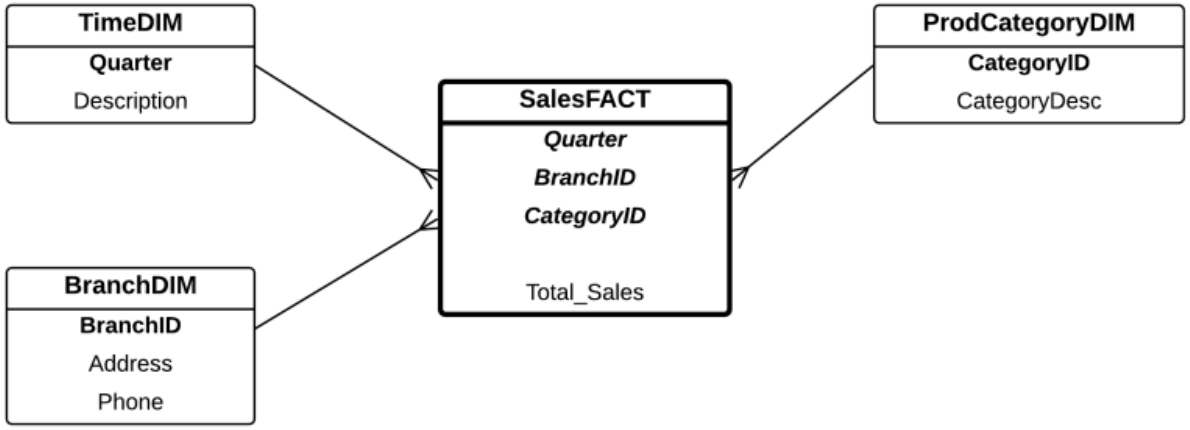


Case Study #2: Sales

- Sales Star Schema
  - **Fact**:
    - Total Sales
  - **Dimensions**:
    - Time
    - Branch
    - Product Category

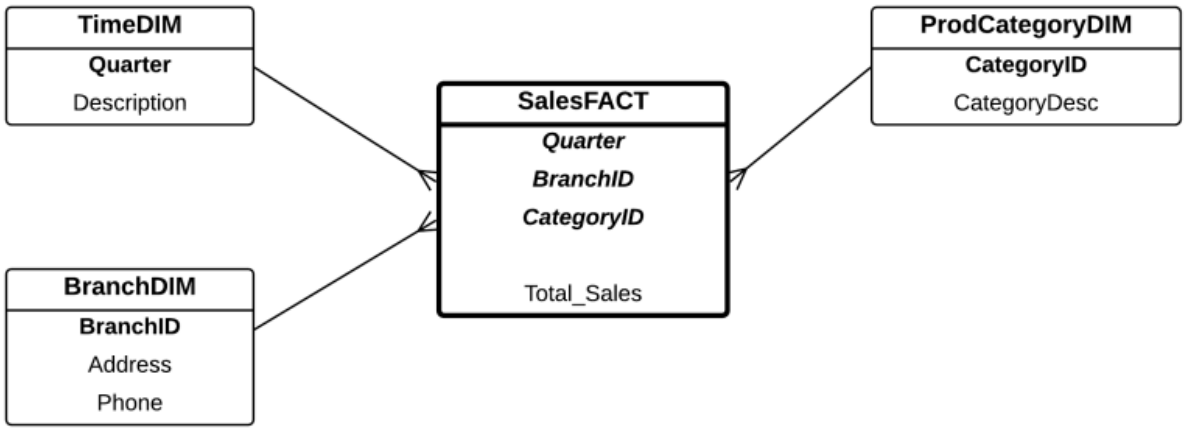
Case Study #2: Sales

Quarter	Description
1	Jan-Mar
2	Apr-Jun
3	Jul-Sep
4	Oct-Dec



Case Study #2: Sales

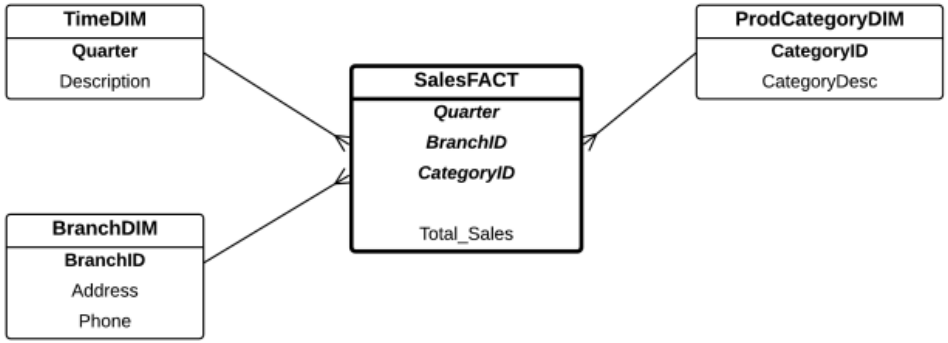
- Sales Star Schema
  - **Fact**:
    - Total Sales
  - **Dimensions**:
    - Time
    - Branch
    - Product Category



Case Study #2: Sales

- To create ProdCategoryDIM:
  - `create table ProdCategoryDim as select * from Category;`

- To create BranchDIM:
  - `create table BranchDim as select * from Branch;`



Case Study #2: Sales

- To create TimeDIM:
  - `create table TimeDim`
    `(Quarter number(1),`
    `Description varchar2(20));`

- To insert the values in TimeDIM:
  - `Insert into TimeDim values (1, 'Jan-Mar');`
    `insert into TimeDim values (2, 'Apr-Jun');`
    `insert into TimeDim values (3, 'Jul-Sep');`
    `insert into TimeDim values (4, 'Oct-Dec');`

Case Study #2: Sales

- To create Temporary Fact for SalesFact:
  - create table TempFact as

```
select
   S.SalesDate,
   B.BranchID,
   C.CategoryID,
   S.TotalPrice
from Branch B, Sales S, Product P, Category C
where B.BranchID = S.BranchID
and S.ProductNo = P.ProductNo
and P.CategoryID = C.CategoryID
and to_char(S.SalesDate, 'YYYY') = '2020';
```

no aggregation function, and no group by

Case Study #2: Sales

- To alter the TempFact:
  -
```
alter table TempFact;
add (Quarter number(1));
```

- To update the TempFact to turn SalesDate into Quarter:
  -
```
update TempFact
set Quarter = 1
where to_char(SalesDate, 'MM') >= '01'
and to_char(SalesDate, 'MM') <= '03';
```

Case Study #2: Sales

- To update the TempFact to turn SalesDate into Quarter:
  -
```
update TempFact
set Quarter = 2
where to_char(SalesDate, 'MM') >= '04'
and to_char(SalesDate, 'MM') <= '06';

update TempFact
set Quarter = 3
where to_char(SalesDate, 'MM') >= '07'
and to_char(SalesDate, 'MM') <= '09';

update TempFact
set Quarter = 4
where Quarter is null;
```

Case Study #2: Sales

- To create SalesFACT:
  - create table SalesFact as
    select
      Quarter,
      BranchID,
      CategoryID,
      sum(TotalPrice) as Total_Sales
    from TempFact
    group by Quarter, BranchID, CategoryID;